RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE-‘A’

Course Code & Title: EC8393 & Fundamentals of Data Structures in C

Practice’s followed: Group Up Group Down

Type of Activity: Collaborative Learning

Topic: Unit-II- Functions, Pointers, Structures and Unions

Date: 08-08-19

Objectives:

O1: To enable the students to develop C programs using function, pointers, structures and unions.

Justification for choosing the particular topic:

Functions, Pointers, Structures and Unions are the very fundamental concept while writing programs in data structure. To make the students strong on the above topic group activity was conducted. This activity helps the students to share their knowledge with their team members and also helps to clarify their doubts.

Steps:

- Group Creation
- Selection of Team Leader
- Writing Questions in the given sheet
- Interchanging of Questions among different groups
- Answering the Question
- Give the justification about answer by team leader

Activity Description:

The class was separated into five groups. Students were insisted on creating unique group names based on the rules of Identifiers. It was quite interesting while creating group names. The group names were

- King_Fisher
- N_U_R_A
- C_oRe_Du_Mpe_d_
- _BIGIL_123
- _adgcompilers

The concept of the activity was actually to frame two questions per team and the questions were interchanged among the groups and the questions were answered after discussing with the group members. At last, it was a discussion time the group leaders were asked to present their answers and justify their answers in front of the class. Finally, answers were discussed.

Reflection critique

Steps taken for address the challenges:

- Permit many students to represent the thoughts and concepts of the topics.
- Advised the students to important of sharing ideas.

Post Implementation:

- From the activity, Leadership Quality of the student improved.
- This activity encouraged the students to share their knowledge with others.

TEAM NAME :_BIGIL_

1.

1) What are the function that supports dynamic memory allocation. with syntax with Eg. 2m

1m

Functions:

1) realloc()
2) calloc()
3) malloc()
4) free()

Syntax:

1) ptr = (int *) malloc (n * size of (int))
2) ptr = (int *) calloc (n, size of (int))
3) realloc (ptr, num)
4) free (ptr)

Calloc ?

Eg:

```
int main()
{
int a[10], i, n, *ptr;
printf("Enter the elements");
Scanf("%d", &n)
ptr = (int *) malloc (n*sizeof(int));
for (i=0; i<n; i++)
{
Scanf ("%d", a[i]);
```

_C_oRe_ Du_Mpe_d_

1) a) Macro is also known as ______, How to declare a Macro processor? (1)

space not allowed

b) Write a C program to print the Id Number, Name, House Number (address), salary, city, Pincode of an employee using structure. (4)

a) define statement (preprocessor Directive)

To declare a Macro preprocessor.

Using # define.

Eg:

```
# include <stdio.h>
# define MIN (a,b)((a)<(b)? (a):(b))
void main()
{
  printf(" min between 10 and 20 is %d", MIN (10,20));
}
```

b)

```
# include<stdio.h>
Struct details
{
    int Idnum;
    char name[10];
    char address[30];
    int salary;
    char city[10];
    int pincode;
};
void main()
{
  printf("enter details");
  printf(" enter Idnum, name, address,
           salary, city, pincode");
  scanf("%d %s %s %d %s %d", &d.Idnum,
        d.name, d.address, &d.salary,
        d.city, &d.pincode);

  printf("%d %s %s %d %s %d", d.Idnum,
         d.name, d.address, d.salary,
         d.city, d.pincode);
}
```

TEAM NAME :_adgcompilers

1. Using Union get biodata for 3 persons (atleast 5 details)

```
# include <stdio.h>
union stu
{
 int age;
 char name[10];
char gender [5];
 int DoB;
int Roll No ;
} s[5];
void main ()
{
printf ("Enter stu details");
for (i=0; i<5 ; i++)
{ Scanf ("%d %s %s %d %d" & s[i].age, s[i].name.
         s[i].gender, s[i], DoB,
```

Relevance of Program Outcomes

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	3	2	2					2	2		1

References

http://archive.wceruw.org/cl1/CL/moreinfo/MI2A.htm
https://teaching.cornell.edu/teaching-resources/engaging-students/collaborative-learning
https://www.thirteen.org/edonline/concept2class/coopcollab/index.html

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